

Bulk Recursion for Integer Multiplication

Large projector blocks, whole residuals, and a pathwise precision guard

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With substantial OpenAI GPT-6 Astra and Codex assistance

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Abstract

We batch rational projector blocks and whole complex residuals in the retained finite networks. A controlled common basis yields contiguous recursive calls; mixed-width moment bounds and a dependency-path precision guard give a saving beyond 2^{-23} . The construction includes integer remainders, bounded row padding, and a corrected Gaussian input error bound. The conclusion is conditional on the pinned upstream multiplication theorem and the retained finite-network and analytic interfaces.

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{6149999}{5 \cdot 10^{13}} > 2^{-23}.$$

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1 Result and retained framework

The model is a fixed finite-alphabet Turing machine with a fixed number of one-dimensional tapes. Section 7.2 gives the rational parameters and strict assembly margins. All constants defining the finite networks, their rational bases, the address radix, and the number of tapes are fixed independently of n . The scalar domains and the address-label fields are distinct. The interchange network carries fixed $\mathbb{F}_3$ payloads and uses rational address matrices. The complex network carries Gaussian dyadic values and uses binary subspace labels.

The finite constructions are those of Zhihao Chen’s PR #7 [8], at commit

6725c6a17b17871a35353fd29157f4ed851bc114.

That contribution supplies the ternary five-subset network, the paired complex producer, their frame schedules, and their finite counts. It builds on Douglas Colkitt’s compact-control framework [2], Bortlesboat’s alignment [3], eumemic’s complex producer and Gaussian resampling [4, 6, 7], and dleen’s retained totals and stage sharing [5]. The upstream manuscript is pinned at OpenAI commit adc7f1241b42e322a6451854ab7e4b4c146bf78a [1].

2 Large rational projectors

2.1 Batching a large-rank projector edge

The usual matrix-shear compiler uses one smaller interchange per rank unit. For a projector of rank close to the ambient dimension, most of these pivot interchanges can instead be one interchange of two contiguous longer fields. The construction below does not gather separated fields or reverse their digits.

A simultaneous rational change of basis. Let $\mathcal{P}$ be a fixed finite family of rational idempotents of dimension m . For $P \in \mathcal{P}$ with $m/2 < a := \text{rank } P < m$, put $r = m - a$. There is one $S \in \text{GL}_m(\mathbb{Q})$ such that, for every such P , the submatrix of SPS^{-1} on its first r rows and last r columns is invertible.

To prove this, first consider one rank- a idempotent. Every rational idempotent is rationally similar to $\text{diag}(I_a, 0)$, by concatenating rational bases of its image and kernel. Put $t = m - 2r > 0$. On the first and last r coordinates take r disjoint rank-one blocks

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

pairing coordinate i with $m + 1 - i$, and take the identity on the middle t coordinates. This is a rational rank- a idempotent whose indicated corner is invertible. Hence the determinant of that corner of SPS^{-1} is not the zero rational function of the entries of S . Clear powers of $\det S$ from its denominator. The product of the resulting nonzero polynomials for the finite family, and $\det S$, is nonzero over $\mathbb{Q}$. A nonzero polynomial over the infinite field $\mathbb{Q}$ cannot vanish at every rational point. This proves simultaneous existence.

The choice is effective: enumerate integer matrices and retain the first invertible one satisfying all the indicated rational determinant tests. This search terminates. The family and m are fixed, so this finite calculation is part of the fixed-machine preparation. Conjugate every terminal and gate matrix by this same S . Common gate matrices, every edge rank, the scalar circuit, and the endpoint

identity are preserved, since $SIS^{-1} = I$. The address prime is then chosen after this conjugation and the triangular factorizations, avoiding their finitely many denominators and invertible diagonal numerators as in the original construction.

Only the edges selected for batching need belong to $\mathcal{P}$. All other matrices can continue to use the original rank-by-rank compiler. In the present finite network, every selected edge is a nonzero scalar multiple of an idempotent: for nested nondegenerate labels $U \subset V$,

$$(P_V - P_U)^2 = P_V - P_U,$$

and reversing the edge only changes its sign. Scalar multiples do not change the pivot locations and are absorbed into the triangular factors.

The contiguous middle pivots. Let $P^2 = P$ have rank $a > m/2$, let $r = m - a > 0$, and assume its first- r -row, last- r -column corner is invertible. Split the ordered coordinates into

$$L = [1, r], \quad M = [r + 1, m - r], \quad R = [m - r + 1, m].$$

The complement $Q = I - P$ has rank r . Write $Q = UV$, where U has r columns, V has r rows, and $VU = I_r$. Such a factorization comes from a basis of $\text{im } Q$. As L and R are disjoint,

$$P_{L,R} = -U_L V_R.$$

Its invertibility makes both square matrices U_L, V_R invertible.

Run the lower-lower elimination in the retained physical order. Its first r pivots occupy the rows L and the columns R , in some permutation. Indeed their corner is invertible, so at each of these steps the next row has a nonzero entry in an as-yet active column of R , which precedes all other candidates in the rightmost-pivot rule. The active Schur complement on rows $M \cup R$ and columns $L \cup M$ has middle block

$$\begin{aligned} P_{M,M} - P_{M,R} P_{L,R}^{-1} P_{L,M} &= I - U_M V_M - (-U_M V_R)(-V_R^{-1} U_L^{-1})(-U_L V_M) \\ &= I_{m-2r}. \end{aligned}$$

Thus the next $m - 2r$ pivots are precisely the diagonal positions (i, i) with $i \in M$. Columns L lie to their left and are cleared by the allowed lower-triangular column operations. The remaining block has rank zero, because the number of isolated pivots is already $r + (m - 2r) = m - r = a$.

Consequently the factorization $P = E_1 \Pi E_2$, with E_1, E_2 invertible lower triangular, has exactly r corner pivots and one contiguous identity block of size

$$t = m - 2r = 2a - m.$$

The same conclusion holds for cP with any fixed nonzero rational c . No assertion that an arbitrary rank- a matrix has this pivot profile is needed; the idempotent identity is what makes the middle block exactly an identity matrix.

One call for the identity block. Subdivide each original contiguous chunk into m consecutive fields of width b . The triangular factors still cost $O(V)$ in the retained later-field affine primitive. Each of the r corner pivots uses one ordinary width- b interchange. For all $i \in M$ together, perform

$$D_i \leftarrow D_i + H_i, \quad (H_M, D_M) \leftarrow (D_M, H_M), \quad D_i \leftarrow H_i - D_i,$$

over $\mathbb{Z}/q^b\mathbb{Z}$. The first and last maps are coordinatewise later-field updates, each using a fixed number of linear scans. The middle map is a single interchange of the two contiguous width- tb chunks H_M, D_M . It preserves the order of their b -digit fields and hence implements all diagonal pivots exactly. The intervening fields are spectators. In particular, addition is componentwise modulo q^b , not addition with carries across the concatenated width- tb chunks.

The edge cost therefore has r recursive calls of width b and one of width tb , in place of a calls of width b , with the same $O(V)$ nonrecursive work. If $a = m$, one may leave the edge in its original m -singleton form. This avoids any same-width recursion. For every batched edge in this paragraph, $0 < t \leq m - 2$, so every child contracts its width by a fixed factor less than one.

The resulting characteristic expression. Let $\mathcal{B}$ be the selected large-rank projector edges, let $r_e = m - a_e$ and $t_e = m - 2r_e$ for $e \in \mathcal{B}$, and retain the total rank sum s . For an exponent $0 < \tau < 1$, the volume-weighted recursive contribution is

$$\mathcal{C}(\tau) = \frac{1}{W} \left[\left(s - \sum_{e \in \mathcal{B}} t_e \right) m^{-\tau} + \sum_{e \in \mathcal{B}} \left(\frac{t_e}{m} \right)^\tau \right]. \quad (2.1)$$

At $\tau = 1$ this is exactly $s/(Wm)$: batching preserves the existing rank deficit. The strict inequality $\mathcal{C}(\tau) < 1$ is the new time-exponent comparison. It uses only the original rank sum and the count of selected ranks; no complete edge-rank histogram is needed. Non-power widths use equal fields of width $\lfloor e/m \rfloor$, at most $m - 1$ separately handled high digits, and the same fixed-tape row recursion. With sufficient rows the nonrecursive work is $O(V)$, so the usual strong induction gives $O(Ve^\tau)$ whenever (2.1) is strict. The separate row and arbitrary-width construction specifies this rounding step and removes the initial row divisibility restriction.

3 The explicit interchange recurrence

3.1 Three geometric rank classes in the retained ternary network

The five-subset producer of PR #7 is retained. Put

$$h = 28, \quad v = \binom{28}{5} = 98280, \quad m = h^3 = 21952, \quad R = 11840940, \quad N = v^3,$$

$$W = 2v^2(v + R), \quad L = 3v^2 \binom{h}{2} (h - 2), \quad s = Wm - N + 2L.$$

This subsection needs the producer's proved width upper bound and frame schedule; it does not enumerate its internal addition graph again. If the producer uses fewer than R roles, insert dummy auxiliary roles with identity gates at the ordinary low and high frames of every invocation. Do this before identifying the first- and third-stage banks. Every dummy role restores its arbitrary scalar, has no decreasing edge, and contributes exactly m to the final rank sum. Thus the stated width and the rank classes below are realized exactly, including any padding.

Write $a = h^{j-1}$ at tensor stage j . With the notation of the retained frame proof, the early auxiliary frame is

$$D_0 = (B \otimes F) \otimes Q, \quad \dim D_0 = (a - 1)h,$$

and the final auxiliary frame is

$$D_1 = (A \otimes F) \otimes Q, \quad \dim D_1 = ah.$$

The two auxiliary banks give the following edges.

1. Each role joining stages one and three passes from a frame of dimension h to a containing frame of dimension $m - h$. This is a projector edge of rank $m - 2h$. There are $B = v^2 R$ such roles.
2. Each stage-two auxiliary role passes from its final frame of dimension h^2 to the full sink frame of dimension m . Its rank is $m - h^2$, again with $B = v^2 R$ copies.

There is one further large class on the data wires. The incoming physical X and Y dimensions at stage j are a and $a - 1$. Their first relevant frames have dimensions $(a - 1)h + 1$ and $(a - 1)h$, respectively. Hence both entrance edges have rank

$$d_j = (a - 1)(h - 1).$$

The remaining growth of each data role in that stage is $h - 1$. These dimensions hold for the reverse middle-stage schedule as well: its low source copy and its target-line injection give precisely the same physical frames. At stage three,

$$d_3 = (h^2 - 1)(h - 1) = 21141 > m/2,$$

and there are $2N$ copies. The special negative source matrix contributes one rank-one edge on each physical X source, as in PR #7. Consequently the complete data rank sum is

$$2N \sum_{j=1}^3 (d_j + h - 1) + N = 2Nm - N.$$

All remaining auxiliary changes inside an invocation have rank at most h , and a stage-two auxiliary entrance has rank $h^2 - h$. Thus the three classes listed above exhaust the ranks exceeding $m/2$, although exhaustion is not needed for the certificate: leaving any other edge unbatched can only weaken the claimed upper bound.

Apply the large-projector batching lemma to these classes. A rank- a_i projector with $a_i > m/2$ uses $m - a_i$ singleton pivot interchanges and one contiguous interchange of $(2a_i - m)$ subfields. The resulting table is

class	copies B_i	rank a_i	singleton pivots	batched width k_i
stage 1–3 join	$v^2 R$	21896	56	21840
stage 2 sink	$v^2 R$	21168	784	20384
stage 3 data entrance	$2v^3$	21141	811	20330

Treat every rank unit outside these three middle blocks as a singleton interchange. The number of singleton calls is therefore exactly

$$S = s - \sum_{i=1}^3 B_i k_i.$$

At a parent width e , the child widths are e/m with multiplicity S and $k_i e/m$ with multiplicity B_i ; all children operate on streams of volume V/W . The finite mixed-width recurrence consequently has normalized moment

$$\Psi(\tau) = \frac{S}{W m^\tau} + \sum_{i=1}^3 \frac{B_i}{W} \left(\frac{k_i}{m} \right)^\tau.$$

At $\tau = 1$ this is $s/(Wm) = 1 - \eta$, where $\eta = 39/520019360$.

The intermediate middle-block certificate. This paragraph records the first batching step. The controlled basis in the following subsection increases its saving from $177/10^9$ to $246/10^9$, which is the value used in the final assembly. Set

$$a_b = \frac{177}{10^9}, \quad \tau = 1 - a_b.$$

Let $r_0 = 1/m$, $r_i = k_i/m$, and

$$w_0 = \frac{S}{Wm}, \quad w_i = \frac{B_i k_i}{Wm}.$$

Then $\sum_i w_i = 1 - \eta$ and $\Psi(1 - a_b) = \sum_i w_i \exp(a_b \log(1/r_i))$. Use the rational logarithm upper bounds

$$\log m < \frac{9997}{1000}, \quad \log \frac{196}{195} < \frac{512}{100000}, \quad \log \frac{14}{13} < \frac{7411}{100000}, \quad \log \frac{21952}{20330} < \frac{768}{10000}.$$

They follow from the positive series

$$\log x = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \quad z = \frac{x-1}{x+1},$$

using sixteen terms and remainder at most $2z^{33}/(33(1-z^2))$; for m , first extract its power of two. For $0 \leq t < 1$, $e^t \leq (1-t)^{-1}$. Calling the four displayed upper bounds ℓ_i , exact rational arithmetic gives

$$\Psi(1 - a_b) < \sum_{i=0}^3 \frac{w_i}{1 - a_b \ell_i} < 1.$$

The strict gap in this last bound is greater than 1.78×10^{-10} . All integers, rational logarithm enclosures, weights and the exact gap are supplied by `scripts/batched_bit_rank_moment.py` and its JSON certificate. Under the accompanying arbitrary-width recurrence lemma, this proves the interchange saving $a_b = 177/10^9$ for the unchanged finite scalar producer. A smaller role upper bound can be inserted directly into the same three-class formula.

3.2 A simultaneous basis with one additional contiguous pivot block

The three-class rank bound can be strengthened while keeping the finite scalar producer unchanged. The additional structure is used only for the stage-two auxiliary sink edges. The same rational change of coordinates preserves the needed corner-rank conditions on the other two large classes.

Put $F = \mathbb{Q}^h$, $V = \mathbb{Q}^H$, $H = h^2$, and $m = hH$. Reorder the tensor factors so that the original third factor is F and the first two factors form V . List coordinates in the order

$$(\alpha, i), \quad 1 \leq \alpha \leq h, \quad 1 \leq i \leq H,$$

with i varying fastest. Write Π_t and Π_q for rank-one idempotents, and Π_U for a rank- h idempotent. The three large projector classes then have the following descriptions:

$$\begin{aligned} A_1 &= I - I_F \otimes \Pi_q - \Pi_t \otimes \Pi_U, & \Pi_q \Pi_U &= \Pi_U \Pi_q = 0, \\ A_2 &= (I_F - \Pi_t) \otimes I_V, \\ A_3 &= (I_F - \Pi_t) \otimes (I_V - \Pi_q). \end{aligned}$$

Their nullities are $2h$, H , and $H + h - 1$, respectively. For the first class, q spans the line associated with $t_B \otimes t_{\pi(A)}$, $U = F \otimes \langle t_A \rangle$, and $t = t_B$. The orthogonality in the bank matching gives the displayed zero products and, in particular, $q \notin U$. The second class is the stage-two sink projection. The third is the stage-three data entrance projection. Transposing any of these matrices preserves its displayed form, with the corresponding dual rank-one lines and subspaces.

Choose $K \in \text{GL}(V)$ and independent matrices $G_i \in \text{GL}(F)$ for $1 \leq i \leq H$. Define

$$T(u \otimes e_i) = G_i u \otimes e_i, \quad S = T(I_F \otimes K).$$

Thus T changes the h first-factor coordinates independently over each of the H multiplicity coordinates. The parameters are rational matrices; S is a change of basis in the finite address-label space, not a tape operation that must be performed on the input streams.

Lemma 3.1 (Simultaneous controlled basis). *For the finite collection of projectors in the retained $h = 28$ motif, one can choose $K, G_1, \dots, G_H$ over $\mathbb{Q}$ so that:*

1. *for every A_1 , the submatrix of SA_1S^{-1} in its first $2h$ rows and last $2h$ columns is nonsingular;*
2. *for every A_3 , the submatrix of SA_3S^{-1} in its first $H + h - 1$ rows and last $H + h - 1$ columns is nonsingular;*
3. *for every A_2 , the submatrix of SA_2S^{-1} in its first H rows and last H columns is diagonal and nonsingular.*

Proof. First consider the third assertion. The factor $I_F \otimes K$ commutes with A_2 . At multiplicity coordinate i , the transformed operator is $G_i(I_F - \Pi_t)G_i^{-1}$. Consequently its first-to-last block is diagonal in i , with diagonal entry

$$[G_i(I_F - \Pi_t)G_i^{-1}]_{1,h}.$$

For a nonzero rank-one projector and $h \geq 2$, this entry is a nonzero rational function of G_i : writing $\Pi_t = t\varphi$ with $\varphi(t) = 1$, it equals $-(e_1^\top G_i t)(\varphi G_i^{-1} e_h)$, and the two factors can both be made nonzero. Thus the required diagonal entries can all be required to be nonzero.

For the first assertion, let $Q_1 = I - A_1$. Its image is the direct sum

$$\text{im } Q_1 = (F \otimes q) \oplus (t \otimes U).$$

Choose a basis matrix B of the h -space U . After applying K , write $q' = Kq$ and $B' = KB$. The nullspace vector parametrized by $u \in F$ and $c \in \mathbb{Q}^h$ has multiplicity- i coordinate

$$G_i(q'_i u + tB'_i c).$$

The first $2h$ coordinate rows therefore evaluate

$$q'_i \ell_i(u) + \ell_i(t)B'_i c, \quad \ell_i = e_1^\top G_i, \quad 1 \leq i \leq 2h.$$

We show that their determinant is not identically zero in the parameters. Choose K so that $q'_i \neq 0$ for $1 \leq i < h$ and the submatrix $[q' \ B']$ in rows $h, \dots, 2h$ is nonsingular. Such a K exists because $[q \ B]$ has column rank $h + 1$ and an invertible linear map can send this ordered independent system to any other one. On the first $h - 1$ rows, choose ℓ_i to be a basis of the annihilator of t . These rows determine u modulo $\langle t \rangle$. On the remaining $h + 1$ rows, choose one fixed functional ℓ with $\ell(t) = 1$. After the first $h - 1$ coordinates of u are determined, these rows form exactly the nonsingular matrix $[q' \ B']$ on the remaining coordinate of u and c . Every prescribed nonzero ℓ_i extends to the first row

of an invertible G_i . This proves that restriction of $\text{Sim } Q_1$ to the first $2h$ coordinates can have full rank.

For the second assertion, the image of $Q_3 = I - A_3$ is

$$\text{im } Q_3 = (\langle t \rangle \otimes V) \oplus (W_t \otimes \langle q \rangle), \quad W_t = \text{im}(I_F - \Pi_t), \quad \dim W_t = h - 1.$$

Since the first summand contains $t \otimes V$, its image under $I \otimes K$ is unchanged. Write a vector after that change as $tv_i + q'_i u$ at multiplicity coordinate i , where $v \in \mathbb{Q}^H$ and $u \in W_t$. The first H coordinate rows give

$$\ell_i(t)v_i + q'_i \ell_i(u), \quad 1 \leq i \leq H.$$

Choose all $\ell_i(t) \neq 0$. They determine the H independent coordinates v_i once u is known. The next $h - 1$ rows are the second rows of G_i at $i = 1, \dots, h - 1$. If $f_i = e_2^\top G_i$, eliminating v_i gives

$$q'_i \left(f_i - \frac{f_i(t)}{\ell_i(t)} \ell_i \right) (u).$$

Choose K with these q'_i nonzero and choose the displayed $h - 1$ functionals to be a basis of the annihilator of t , restricted to W_t . This is possible because $F = \langle t \rangle \oplus W_t$: take each $\ell_i(t) = 1$, take the relevant f_i to annihilate t , and complete these independent first two rows to G_i . Thus restriction to the first $H + h - 1$ coordinates can also have full rank.

Apply the same two arguments to the transposed projectors at the last coordinate rows. This is a condition within the same parameter family, because

$$S^{-\top} = T^{-\top}(I_F \otimes K^{-\top}),$$

and $T^{-\top}$ acts independently by $G_i^{-\top}$. They show that the corresponding last-coordinate restrictions of the transformed dual nullspaces can each have full rank.

To combine the requirements, express every rank condition as a nonzero minor after clearing the denominators of the parameter matrices and their inverses. Each is a polynomial that is not identically zero, by the constructions just given. Their finite product, together with the invertibility conditions on K and all G_i , is still a nonzero polynomial. A nonzero rational polynomial cannot vanish at every rational tuple, so one rational choice meets all conditions simultaneously. The individual witness choices above establish that the polynomials are nonzero; they need not themselves meet all the other conditions.

Finally, for a rank- r nullspace projector $Q = UV$ with $VU = I_r$, its first- r by last- r submatrix after conjugation factors as the product of the two square restriction matrices just considered. Both are nonsingular. The corresponding block of $I - SQS^{-1}$ is its negative, since these first and last index sets are disjoint. This proves the first two assertions. The third has already been proved directly. $\square$

The argument is constructive in the finite-data sense required by the network interface: enumerate rational parameter matrices and retain the first choice passing the finitely many specified determinant tests. The lemma proves termination. All parameters and tests depend only on the fixed motif, not on the eventual multiplication input size. Conjugate every terminal and gate matrix by this one S . Their endpoint differences remain I_m , common gate matrices remain common, and all edge ranks are preserved.

The improved stage-two pivot profile. The lower-triangular elimination for a large projector first eliminates its nonsingular corner block. For A_2 , this block is already diagonal and its H pivots lie in matching physical order in two contiguous sets of fields. After fixed diagonal normalizations, they are one $H \times H$ identity block. The Schur-complement argument in the large-projector lemma leaves the middle identity block of size $m - 2H$. Hence a stage-two sink edge is implemented by two contiguous child interchanges, of widths

$$Hb \quad \text{and} \quad (m - 2H)b,$$

with no singleton recursive calls on its corner. Its rank remains $H + (m - 2H) = m - H$, so the rank sum at exponent one is unchanged. The first and third edge classes retain their previous singleton-corner and contiguous-middle implementations by the first two parts of the lemma.

Exact strengthened moment. Let $B = v^2 R$ and let S_0 be the singleton count from the three-class certificate. Replace S_0 by $S_0 - BH$ and add B children of width H/m . For $h = 28$ the new singleton count is

$$S_1 = 105555640096680883200.$$

The normalized child widths and their rank-mass weights are respectively

$$\left(\frac{1}{21952}, \frac{195}{196}, \frac{13}{14}, \frac{10165}{10976}, \frac{1}{28} \right)$$

and

$$\left(\frac{10841531}{520019360}, \frac{12827685}{26000968}, \frac{855179}{1857212}, \frac{20865}{2736944}, \frac{65783}{3714424} \right).$$

Use the four preceding rational logarithm bounds and additionally $\log 28 < 33323/10000$. With

$$a_b = \frac{246}{10^9}, \quad \tau = 1 - a_b,$$

the same bound $e^x \leq (1 - x)^{-1}$ gives a strict rank-moment gap exceeding 4.76×10^{-11} . The exact rational calculation is in `scripts/controlled_bit_rank_moment.py`. This proves the stronger interchange saving using the unchanged PR #7 scalar producer and the same arbitrary-width recurrence interface.

4 Integer widths and bit-stream rows

4.1 Integer widths, bounded row padding, and mixed contractions

The batched residual construction uses a fixed finite list of child sizes, not only powers of one radix. The following implementation supplies the corresponding integer-width recursion. All constants in this subsection are fixed finite circuit constants, independent of the input length.

A general moment condition. Let $m, W \geq 2$ be fixed. At one circuit invocation, let the list $t_1, \dots, t_J$ contain integers in $[1, m - 1]$. A child of type i has width $t_i f$, where $f = \lfloor e/m \rfloor$, and operates on exactly $1/W$ of the parent's logical volume. Repetitions in the list represent separate calls. An ordinary rank-one call has $t_i = 1$; a batched block has the corresponding larger value of

t_i . Full-rank edges may retain their original m rank-one calls, so no same-width recursive call is needed. Put

$$\Psi(x) = \frac{1}{W} \sum_{i=1}^J (t_i/m)^x, \quad t_* = \max_i t_i < m.$$

If $\Psi(\tau) < 1$ for some $0 < \tau < 1$, then the normalized recurrence

$$F(e) \leq \frac{1}{W} \sum_{i=1}^J F(t_i \lfloor e/m \rfloor) + C, \quad F(e) = O(1) \quad (1 \leq e < m),$$

has $F(e) = O(e^\tau)$ for every positive integer e . Indeed, substituting $F(u) \leq Au^\tau$ into an internal call gives at most $A\Psi(\tau)e^\tau + C$. Choose A to cover the finite base range and $C/(1 - \Psi(\tau))$. Floors can only decrease the right-hand side.

A finite integer depth bound. Define

$$D(e) = 0 \quad (e < m), \quad D(e) = 1 + D(t_* \lfloor e/m \rfloor) \quad (e \geq m).$$

This is an integer algorithm: repeatedly replace e by $t_* \lfloor e/m \rfloor$ until it is below m . The function D is nondecreasing, every child has depth bound at most $D(e) - 1$, and $D(e) = O(\log(2e))$, because each internal replacement is at most $(t_*/m)e$. A row count divisible by $W^{D(e)}$ is therefore sufficient for every root-to-leaf path. At each invocation write the row index as $Wg + w$ and split by w . Every child has complete within-row ranges, logical volume exactly V/W , and a remaining row count divisible by $W^{D(e)-1}$. No padding occurs inside this recursion.

Bit interchange with a remainder. At an integer-width call write $e = mf + r$, $0 \leq r < m$. If $f = 0$, interchange the bounded number of digits individually. Otherwise separate the high r digits of each of the two target chunks. With G denoting all intervening spectators, the layout is

$$(H^+, H^-, G, D^+, D^-).$$

First exchange corresponding digits of H^+, D^+ and move H^+ just before H^- . The layout becomes (D^+, H^+, H^-, G, D^-) . Apply the finite circuit to the two length- mf chunks H^-, D^- . Finally move H^+ back immediately before the final H^- , obtaining

$$(D^+, D^-, G, H^+, H^-).$$

All other fields are preserved. These moves use fewer than a fixed multiple of m single-digit operations and cost $O(V)$. The main chunks split into m equal width- f fields, so every rational shear and every batched contiguous-block shear has exactly the required field shape. A child retains the unused main digits and all remainders as spectators. Thus the preceding moment recurrence applies literally at every integer width; there is no per-level numerical padding of either chunk range.

Supplying the bit rows from existing digits. For an arbitrary width e , compute $D(e)$ and choose the least nonnegative integer ρ such that

$$q^{2\rho} \geq W^{D(e)}.$$

Both numbers can be constructed by fixed-radix integer arithmetic. Since $D(e) = O(\log e)$, we have $\rho = O(\log e)$. For sufficiently large e , $2\rho < e$. The exceptional set of smaller widths is fixed and finite; individual-digit interchange there is absorbed by the asymptotic constant.

Exchange the high ρ digits of the two chunks, then gather their two high parts into one row field before both remaining chunks, exactly as in the preceding layout operation. This row field has range $q^{2\rho}$. Within each original prefix, pad its range to

$$R' = W^{D(e)} \left\lceil q^{2\rho} / W^{D(e)} \right\rceil < 2q^{2\rho}.$$

Every added row includes the entire remaining address set. Recurse once at the remaining width $e - \rho$, which has depth bound at most $D(e)$; then remove the added rows and undo the gathering move. Each completed invocation preserves its row index, so padded zero rows are zero again when they are removed.

The possibly large exponent in $W^{D(e)} = e^{O(1)}$ does not multiply the logical volume by that polynomial. The row digits were already part of the two input chunks. Only rounding the row count adds data, by a factor strictly below two, once at the outer boundary. Exchanging and gathering $O(\log e)$ high digits costs $O(V \log(2e))$, which is $O(Ve^\tau)$. The radix conversion from binary chunk ranges to the fixed radix q retains the original constant factor, less than q^2 . Consequently the batched construction is an arbitrary-width interchange procedure with the asserted exponent τ and no additional power of its volume or width.

Fixed tapes and descriptors. Use the original depth-first scheduler: park only the other role streams at the active stack top, run the child on one complete role stream, and restore the parked streams on return. The number of child types is fixed, so parking, copying, cleanup and the finite arithmetic transforms contribute $O(V)$ per node. Before descending, retain the two current chunks and merge consecutive spectator fields; the descriptor has a fixed number of field intervals. Its integer lengths have $O(\log V_0)$ bits. All descendants retain the complete ranges of both root main chunks, even when most digits have become spectators. Their volume is therefore at least $q^{2(e-\rho)}$ in the outer construction, while $j \log W = O(\log e)$ at depth j . Hence $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. Polynomial descriptor work is absorbed by $O(V_j)$, as in the original fixed-tape proof. The larger fixed constant in the depth bound creates no new tape and no growing alphabet.

5 Complex residuals and arithmetic depth

5.1 The weighted arithmetic-path budget

The precision analysis uses the length of an arithmetic dependency path through the framed complex network. We first measure this path in directional rank units, before grouping some units into bulk children.

A weighted path bound. Use the retained triple-based complex motif on h points, with $m = h^3$. This includes the paired complex producer and its shared first- and third-stage auxiliary banks. Regard every scalar gate as a map from saved inputs to its outputs. All of its inputs and outputs have one common subspace label. A directed dependency path can therefore enter a gate on one physical wire and leave on another without changing label dimension at the gate itself.

For a network edge a , let U_a^- and U_a^+ be its labels and write $r_a = |\dim U_a^+ - \dim U_a^-|$. The edge implementation has exactly r_a recursive calls, each implementing one directional C tensor on e/m

selected axes. Binary changes of coordinates before and after these calls permute coefficients and add no arithmetic depth. An inverse direction is implemented using one forward child and its sign and fourth-root wrappers, so it has the same recursive-call count.

If P is a directed scalar dependency path, its common gate labels give the exact identity

$$\sum_{a \in P} r_a = \dim U_{\text{end}} - \dim U_{\text{start}} + 2 \sum_{\substack{a \in P \\ \dim U_a^+ < \dim U_a^-}} (\dim U_a^- - \dim U_a^+).$$

The endpoint difference is at most m .

Every side-producer edge is increasing. This follows from nested source labels for the nonconstant middle mixer, complemented labels in its reverse-direction counterpart, and constant early and late frames. The data-wire and interstage edges are increasing as in the retained motif. The stage-sharing joins are also increasing. The only decreasing edges are the retained central returns, each of dimension loss h .

There are three tensor stages. Within a stage, the v^2 invocations use disjoint data and auxiliary wires, and each invocation has a single central-return frontier. Equivalently, its point and total central wires all return between the same two blocks of the transparent schedule. A directed path crosses that frontier at most once. Gates acting on different invocations do not connect their wires. Thus a path encounters at most one central return per stage, even though the machine executes all invocations sequentially on its fixed tapes. Sharing stage-1 and stage-3 banks only links their existing stage boundaries and creates no additional return. Consequently

$$\sum_{a \in P} r_a \leq q := m + 6h. \quad (5.1)$$

The bound also applies to a path ending at an intermediate gate or inside an edge implementation: complete its current edge, and then any suffix to a terminal vertex. All weights are nonnegative.

From a scalar path to a coefficient path. A coefficient dependency path in the actual tape algorithm projects to a directed path of the scalar network: inside a scalar gate it follows one of the saved operands that contributes to the chosen output; inside an edge it stays on the edge's role stream. A child can permute or combine addresses within that role, but cannot move a value into a different parent role. The row split, parking, padding and reassembly operations do not create arithmetic dependencies between distinct parent roles. This projection justifies using (5.1) for the implemented recursion, rather than merely for a drawn finite circuit.

5.2 A whole-residual complex child

The complex phase interface admits a further batching step. On an edge with residual dimension a , the existing binary change of basis puts an orthonormal residual basis into the first a slots. If each slot has f selected axes, those first a slots form one contiguous interval of af axes. Instead of a child calls on f axes, use one child on that interval.

The directions need not all have the same sign. A basis vector's weight modulo four may make its directional kernel an inverse even on an increasing edge. Let B be the set of negative residual slots and let Z_B multiply an address by -1 precisely when the parity of its selected bits in those slots is odd. The exact identity $C^{-1} = -iZCZ$ gives

$$\bigotimes_{j=1}^a (C^{\varepsilon_j})^{\otimes f} = (-i)^{f|B|} Z_B C^{\otimes af} Z_B, \quad \varepsilon_j \in \{1, -1\}.$$

Thus one forward child, two address-sign passes and one fixed fourth-root phase implement the mixed signs. The same binary basis changes before and after the child are retained. The wrappers cost $O(V)$: their sign is computed from the current address descriptor, whose polynomial work per record is absorbed by the existing superpolynomial record length. They introduce at most four elementary coefficient operations per value and no further recursive call.

We batch only the two specified auxiliary edge classes of ranks

$$a_1 = m - 2h, \quad a_2 = m - h^2.$$

All other edges continue to use their old one-direction children. The first class is the increasing join between a completed stage-1 auxiliary bank, at dimension h , and its matched stage-3 bank, at dimension $m - h$. The second is the stage-2 auxiliary exit from dimension h^2 to the full terminal frame. These are complete auxiliary banks, including the central roles when those are shared. If a width bound requires padding, add dummy auxiliary roles with identity scalar gates at the ordinary low and high frames before matching the first- and third-stage banks. Their labels only increase, their arbitrary scalar values are restored, and both counted auxiliary classes are then present for every padded role. The changed volume-weighted time expression is consequently

$$\frac{1}{W} \left[(s - n_1 a_1 - n_2 a_2) m^{-\sigma} + n_1 \left(\frac{a_1}{m} \right)^\sigma + n_2 \left(\frac{a_2}{m} \right)^\sigma \right],$$

where n_1, n_2 count the chosen edges. It has value $s/(Wm) < 1$ at $\sigma = 1$. Strict comparison below one supplies the new complex recursion exponent. The separate mixed-width row construction handles $f = \lfloor e/m \rfloor$, the at most $m - 1$ individually processed remaining axes, and a row reservation sufficient for the longer but still $O(\log d)$ recursion.

5.3 A depth guard for the unequal complex children

The dependency-path argument of Section 5.1 still bounds the sum of edge ranks on one scalar path by

$$q = m + 6h.$$

For $h = 28$ its relevant constants are

$$m = 21952, \quad q = 22120, \quad a_1 = 21896, \quad a_2 = 21168.$$

Both selected ranks exceed $q/2$. Therefore every arithmetic dependency path encounters at most one selected bulk edge. At a call on e axes, put $f = \lfloor e/m \rfloor$. Ignoring the local additive operations, the path has one of the following child-depth bounds:

$$qA(f), \quad (q - a_1)A(f) + A(a_1 f), \quad (q - a_2)A(f) + A(a_2 f).$$

The weight budget accounts for both increasing and decreasing frame edges and for paths that switch wires at a scalar gate. Inverse-child and endpoint sign corrections add no child. Individually processing the fewer than m leftover axes adds at most $8m$ elementary operations, absorbed into the local constant below.

Take $\rho = 6/5$. The following three normalized path expressions have a rational common upper bound strictly below one:

$$\max \left\{ \frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m} \right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m} \right)^\rho \right\} < \bar{\theta} := \frac{999}{1000}. \quad (5.2)$$

Here is an entirely rational verification. First $160000^5 < m^6$, so $m^\rho > 160000$. For $0 < t < 1$, Taylor's formula with its second derivative bounded on $[0, t]$ gives

$$(1-t)^{6/5} \leq 1 - \frac{6}{5}t + \frac{3}{25} \frac{t^2}{1-t}.$$

Indeed its second derivative is $(6/25)(1-t)^{-4/5} \leq (6/25)/(1-t)$. Putting $t = (m - a_i)/m$ gives upper bounds, in the displayed order,

$$\frac{553}{4000}, \quad \frac{47817969}{47897500}, \quad \frac{1213697}{1260000},$$

all strictly below 999/1000.

Stopping and exact arithmetic. Retain a global stopping threshold d^β , with $0 < \beta < 1$. For sufficiently large d , this threshold exceeds $2m$. A leaf reached from an internal call has width at least $d^\beta/(2m)$: every child has width at least $f = \lfloor e/m \rfloor \geq e/(2m)$. Put

$$E = 64(W + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1).$$

The exact finite inequality $36W^3 + 4s + 4W + 8m + 4 < E$ covers all local scalar operations, the new leftover-axis work and the retained endpoint and inverse wrappers. For every descendant width e reached before or just after stopping, strong induction gives

$$A(e) \leq C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho. \quad (5.3)$$

At a leaf, use

$$8e \leq 8(2m)^{\rho-1} d^{-\beta(\rho-1)} e^\rho \leq 16m d^{-\beta(\rho-1)} e^\rho.$$

At an internal call, child widths are at most their stated fractions of e , so (5.2) bounds their total by $\bar{\theta} C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho$. Since $e \geq d^\beta$, the remaining gap is at least $(1 - \bar{\theta}) C_{\text{dep}} d^\beta \geq E$. This proves the induction. A root that is already a leaf separately satisfies $A(e) \leq 8d^\beta$, which is smaller than the final bound below.

For rational $\zeta > 0$, set

$$C_1 = \rho - (\rho - 1)\beta + \zeta, \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

The root bound from (5.3) is $C_{\text{dep}} d^{\rho-(\rho-1)\beta}$. This covers the direct arbitrary-width root construction; it also covers every retained base- m piece. Even retaining the older bound of $m(1 + 1/\zeta)d^\zeta$ such pieces, their concatenation, all reserved axis kernels, the outer phases and the b^D conversion have depth at most $C_0 d^{C_1}$. The original elementary-depth induction therefore supplies exact denominator and magnitude guards of $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits. The width remains $O(p)$ when $\epsilon C_1 < 1$.

In particular, $\beta = 1/1000$ and $\zeta = 1/10000$ give

$$C_1 = \frac{11999}{10000}, \quad 1 - \frac{7999999}{16000000} C_1 = \frac{64008011999}{160000000000} > 0.$$

Thus bulk complex children preserve the near-half dimension exponent while allowing the improved complex time exponent to be used with almost no loss at the stopping leaves.

6 The complete layer on mixed widths

6.1 The complex layer with unequal recursive axis counts

Apply the same integer-depth construction to the complex circuit's list of child sizes. For a selected residual of rank $a < m$, the residual orthonormal basis places its nontrivial coordinate slots first. Its a consecutive slots of f selected axes form one contiguous block of af axes. The residual factors can have signs $\varepsilon_v \in \{1, -1\}$. For their negative slot set B , the retained identity $C^{-1} = -iZCZ$ gives, after the residual basis map,

$$\prod_{v=1}^a (C^{\varepsilon_v})^{\otimes f} = (-i)^{|B|} Z_B C^{\otimes af} Z_B.$$

Here Z_B applies Z on the f axes of each negative slot. Thus one recursive application of $C^{\otimes af}$, two address-sign passes, and one exact phase replace the a separate recursive calls; the basis maps before and after the operation are unchanged. The sign descriptors have polynomial size in the global parameters and fit the existing per-record overhead allowance. Other residuals retain their original rank-one calls. Let Ψ_c and D_c denote the resulting moment and depth functions. In particular, the two bulk classes of the current motif have

$$a_1 = m - 2h = 21896, \quad a_2 = m - h^2 = 21168 \quad (m = 21952, h = 28),$$

so every recursive call is a strict contraction.

Complete rows and compact fields. Keep the original global p, d, K, G, H, q_F, q_B for compact control, and replace only the row reservation by

$$k_0 = D_c(d), \quad q_0 = \lceil \log_2 W \rceil k_0, \quad Q_0 = q_0 + q_F + q_B.$$

This conservative integer choice satisfies $2^{q_0 K} \geq W^{k_0}$ and $q_0 = O(\log(2d))$. If the available number D of selected axes is at most Q_0 , apply them individually. Otherwise process the first $q_0 + q_F$ and last q_B axes individually and use the remaining consecutive middle block for the recursive circuit. The row field consists only of the first q_0 chunks; neither compact field is included in it. Pad its count to the next multiple of W^{k_0} , by a factor below two, once before recursion. Every child therefore has all compact and spectator ranges and exactly $1/W$ of its parent's volume, with enough row divisibility for its remaining depth.

The reserved-axis bound is unchanged:

$$Q_0 = O(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1).$$

In particular, the larger fixed row-depth constant does not change any power of d or p in preprocessing. The exact bit interchange called inside a compact operation supplies its own rows from its own target chunk digits; its internal row bound is not an additional reservation of selected complex axes.

Integer axis counts. At a complex call of width $e = mf + r$, first apply the $r < m$ leftmost selected kernels individually and treat those axes as completed spectators. The remaining mf axes are consecutive and split into the required m equal slots. The extra work is $O(V)$ per node. The individual kernels on removed axes commute with the completed tensor map on the other axes, so this produces the desired map on all e axes. The root active block can be processed directly; its base- m expansion is unnecessary.

Stopping and weighted-tree bounds. Fix rational $0 < \beta < 1$ and stop at $e < d^\beta$, with normalized leaf cost $O(e)$. Suppose $\Psi_c(\sigma) < 1$ for $0 < \sigma < 1$. Each node's normalized volume weight is W^{-j} at depth j . Its children have total σ -weighted width at most $\Psi_c(\sigma)$ times the parent's weighted width. Therefore

$$\sum_{\text{leaves } u} W^{-\text{depth}(u)} e_u^\sigma \leq e_{\text{root}}^\sigma.$$

Since $e_u < d^\beta$, total normalized leaf work is at most

$$O(d^{\beta(1-\sigma)} e_{\text{root}}^\sigma) = O(d^{\sigma+\beta(1-\sigma)}).$$

This is a bound for a tree with unequal child widths; no common leaf depth is assumed.

Compact operations cost $O(G^\tau e^\tau + 1)$ per current logical volume. If $\sigma \leq \tau$, monotonicity gives $\Psi_c(\tau) \leq \Psi_c(\sigma) < 1$, and summing the τ -weighted widths by depth bounds all internal work by $O(G^\tau d^\tau)$. If $\sigma > \tau$, every internal width is at least d^β , so $e^\tau \leq d^{\beta(\tau-\sigma)} e^\sigma$; summing the strict σ -moment gives $O(G^\tau d^{\sigma+\beta(\tau-\sigma)})$. Thus in both cases, with

$$\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\},$$

the complete normalized layer cost is

$$O((\log p)^3 [d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c, 0\}} + 1]).$$

The displayed logarithmic factor conservatively retains the existing layout and descriptor allowances. Hence the original strict conditions

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda'$$

continue to imply the complete $O(Vd^{\lambda'})$ layer bound.

Compatibility with the new dependency guard. The separate dependency-path estimate uses $q = m + 6h = 22120$, observes that one path crosses at most one selected bulk residual, and bounds its ρ -moment by

$$\max \left\{ \frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m} \right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m} \right)^\rho \right\} < \frac{999}{1000} \quad \left(\rho = \frac{6}{5} \right).$$

Integer floors only shrink these children. For sufficiently large d , an internal parent has $e \geq d^\beta \geq 2m$, hence every recursive child has width at least $\lfloor e/m \rfloor \geq d^\beta/(2m)$ whenever it becomes a stopping leaf. The at most $m - 1$ individually processed remainder axes at each internal node belong to its fixed local overhead; add their bounded coefficient contribution to the guard's fixed constant E . They are not extra arbitrarily small stopping leaves. Thus the separately established guard exponent

$$C_1 = \rho - (\rho - 1)\beta + \zeta = \frac{6}{5} - \frac{\beta}{5} + \zeta$$

applies to this integer-width implementation. All stopping tests can be performed by the existing rational-power integer comparison.

7 Exact conditional assembly

7.1 An exact complex exponent after bulk recursion

Keep the paired complex producer of PR #7. Its constants are

$$h = 28, \quad v_c = \binom{28}{3} = 3276, \quad m = 21952, \quad R_c = 93838.$$

There are $R_c + h + 1 = 93867$ auxiliary roles per invocation, including the original central roles. Write

$$\begin{aligned} B_c &= v_c^2(R_c + h + 1) = 1007397164592, \\ W_c &= 2v_c^2(v_c + R_c + h + 1) = 2085111546336, \\ L_c &= 3v_c^2h(h + 1) = 26143580736, \quad s_c = W_cm - 2v_c^3 + 2L_c = 45772350635112192. \end{aligned}$$

The two selected whole-residual classes have ranks $a_1 = 21896$ and $a_2 = 21168$, each with B_c copies. The number of remaining singleton children is

$$S_c = s_c - B_c(a_1 + a_2) = 2389799139122304.$$

Their time moment is

$$\Xi(\sigma) = \frac{S_c}{W_cm^\sigma} + \frac{B_c}{W_c} \left[\left(\frac{a_1}{m} \right)^\sigma + \left(\frac{a_2}{m} \right)^\sigma \right].$$

At $\sigma = 1$, this equals $1 - \eta_c$ with $\eta_c = 5/12693352$.

Take

$$a_c = \frac{7}{10^7}, \quad \sigma = 1 - a_c.$$

The same positive logarithm series used in the bit certificate gives

$$\log m < \frac{9997}{1000}, \quad \log \frac{m}{a_1} = \log \frac{392}{391} < \frac{2555}{10^6}, \quad \log \frac{m}{a_2} = \log \frac{28}{27} < \frac{36368}{10^6}.$$

Let these upper bounds be ℓ_0, ℓ_1, ℓ_2 and put

$$w_0 = \frac{S_c}{W_cm}, \quad w_i = \frac{B_c a_i}{W_cm} \quad (i = 1, 2).$$

The exact rational comparison is

$$\Xi(1 - a_c) < \sum_{i=0}^2 \frac{w_i}{1 - a_c \ell_i} < 1.$$

The gap in the last inequality is

$$\frac{24132873235669673777179671369131}{1525632551364197359955324563293421369131} > 1.58 \times 10^{-8}.$$

This certifies the stronger complex exponent with the unchanged finite producer and the new whole-residual recursion.

7.2 Final multiplication parameters and margins

Use the controlled-basis projector-batched interchange saving $a_b = 246/10^9$ and the preceding complex saving $a_c = 7/10^7$. The complete rational parameter choice is

$$\begin{aligned}\tau &= 1 - \frac{246}{10^9}, & \sigma &= 1 - \frac{7}{10^7}, & \epsilon &= \frac{7999999}{16000000}, \\ c &= 1, & \beta &= \frac{1}{1000}, & \delta &= \frac{1}{10^{10}}, \\ \zeta &= \frac{1}{10000}, & C_1 &= \frac{11999}{10000}, \\ \lambda &= 1 - \frac{246}{10^9} + \frac{1}{10^{16}}, & \lambda' &= 1 - \frac{246}{10^9} + \frac{2}{10^{16}}, \\ \kappa &= \frac{6149999}{5 \cdot 10^{13}}.\end{aligned}$$

The value $c = 1$ is allowed by the compact-control construction. Its reservation exponent $\max\{1 - c, 0\}$ is zero, while $1 - \epsilon(1 + c) = 1/8000000 > 0$ still ensures $K = o(\ell)$ for $K = d$. The larger fixed constants in the mixed-width row reservations do not alter any power of p .

Since $\sigma < \tau$, the compact-control internal exponent is τ . Its leaf exponent is

$$\sigma + \beta(1 - \sigma) = 1 - \frac{6993}{10^{10}} < \tau.$$

The strict gaps of 10^{-16} between τ, λ, λ' absorb the fixed powers of $\log p$ in the layer construction. The new coefficient guard satisfies

$$1 - \epsilon C_1 = \frac{64008011999}{160000000000} > 0.$$

The fast Gaussian construction retains $\alpha = \lfloor \sqrt{b/(8d)} \rfloor$ and $\gamma = 2d\alpha^2 \leq b/4$. Here $1 - 2\epsilon = 1/8000000 > 0$, so its eventual hypotheses $b \geq 64d^2$, $\alpha^2\theta_i \geq 1$ and the prime-interval growth condition all continue to hold. The numerical maps, their target errors, the final rounding step and the input/output format are unchanged. For the retained chirp input generator, use $F = 2^{\lceil 1.14\alpha^2 \rceil}$ to bound both magnitude and shifted absolute error; the corrected block argument still uses $P = 34p$.

The seven margins below one in the assembly cost table are

$$\begin{aligned}g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), & g_3 &= \epsilon(1 - \lambda'), \\ g_4 &= (1 - \epsilon)(1 - \tau), & g_5 &= 1 - \delta - 2\epsilon, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon.\end{aligned}$$

Their exact values are

margin	value
g_1	$1/8000000$
g_2	$983999877/8000000000000000$
g_3	$9839998762000001/80000000000000000000$
g_4	$984000123/8000000000000000$
g_5	$1249/10000000000$
g_6	$312500039/625000000$
g_7	$7999999/16000000$

All twenty-nine parameter constraints are strict. The minimum final margin is $G = g_3$, and

$$G - \kappa = \frac{362000001}{800000000000000000000} > 0.$$

This last fixed positive gap absorbs the remaining logarithmic factors. The resulting conditional multiplication bound is

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{6149999}{5 \cdot 10^{13}} = 1.2299998 \times 10^{-7} > 2^{-23}.$$

The exact comparisons are reproduced by `scripts/batched_network.py` and recorded in `certificates/batched-network.json`.

8 Precision correction to the retained Gaussian interface

The retained fast Gaussian construction generates $2^{-B}e^{\pi\alpha^2\beta_j^2}$ at precision P , where $B = \lceil 1.14\alpha^2 \rceil$, then shifts by B and multiplies by the disk-grid input u_j . The exponential interface gives absolute error at most 2^{1-P} before the shift. Thus the error after shifting is controlled by 2^{B+1-P} , not by a constant multiple of $2^{-P}e^{\pi\alpha^2/4}$ uniformly in α .

Set $F = 2^B$ in the retained chirped-correlation block lemma. Since $|\beta_j| \leq 1/2$ and $\pi/(4\log 2) < 1.14$, this bounds the exact input magnitude. Including the product and rounding, it also gives the required input accuracy $|\tilde{f}_j - f_j| \leq 2^{4-P}F$. The distinction matters: $2^B/e^{\pi\alpha^2/4}$ is unbounded, so the smaller exact-magnitude bound alone does not justify this accuracy.

No additional asymptotic precision is needed. For the inherited integer parameters $p > 100$ and $2 \leq \alpha < \sqrt{p}$, let $L_A = 3\lceil \sqrt{p}/(2\alpha) \rceil + 1$ and $\Gamma = 29p$. Then $L_A \leq p$, $\lceil \log_2 L_A \rceil \leq p - 1$ and $B \leq 1.14p + 1$. The block lemma requires

$$P \geq p + \Gamma + \lceil \log_2(FL_A) \rceil + 11.$$

Its right side is at most $32.14p + 11 < 34p$, so the retained $P = 34p$ satisfies the corrected error budget. The packing widths remain $O(p)$, and the target errors, tape-time exponents and final rounding guarantee are unchanged. The combined manuscript patch includes this correction directly in the retained Gaussian proof.

9 Constructive algorithm specification

9.1 Constructive algorithm for batched integer-width interchange

The following pseudocode specifies how the fixed finite circuit becomes an interchange procedure. It uses the retained single-digit streaming, ordered-affine streaming, finite-circuit transfer, and depth-first tape primitives.

One-time finite preparation in the controlled tensor basis. Start from the prime-field five-subset circuit, its corrected role permutation, and its all-role rational endpoint matrices. The payload alphabet is $\mathbb{F}_3$; the integer-address radix below is a separate fixed odd prime q .

Put $H_0 = h^2 = 784$ and reorder the finite label-space tensor factors so that the original third factor is first. Write the resulting space as $F \otimes V$, where $\dim F = h$ and $\dim V = H_0$, with coordinates (α, i) , $1 \leq \alpha \leq h$, $1 \leq i \leq H_0$, and i varying fastest. The controlled basis of Lemma 3.1 has the form

$$T(u \otimes e_i) = G_i u \otimes e_i, \quad S = T(I_F \otimes K),$$

where $K \in \text{GL}_{H_0}(\mathbb{Q})$ and the $G_i \in \text{GL}_h(\mathbb{Q})$ are chosen independently. This choice applies to the assigned finite matrices; it introduces no extra change of coordinates on the tape streams.

PREPARE_FINITE_DATA(circuit):

```

Reorder the finite tensor coordinates as (third, first-and-second).
Enumerate rational invertible K and rational invertible G_i.
Form T(u tensor e_i) = G_i*u tensor e_i; S = T*(I tensor K).
Retain the first choice satisfying the finite controlled-basis tests:
    A1: its first 2h by last 2h corner is nonsingular.
    A3: its first H0+h-1 by last H0+h-1 corner is nonsingular.
    A2: its first H0 by last H0 corner is diagonal and nonsingular.
Conjugate every terminal and gate matrix by this same S.
For each edge, form A = matrix_at_head - matrix_at_tail.
If A belongs to A1 or A3, of rank a:
    r = m-a; t = 2*a-m.
    Store lower-triangular factors A = E1*Pi*E2.
    Store r corner pivots and one middle identity block of t slots.
If A belongs to A2:
    Store lower-triangular factors A = E1*Pi*E2.
    Store the aligned corner identity block of H0 slots and
        the middle identity block of m-2*H0 slots.
    Store no singleton corner pivots.
Otherwise:
    Store the ordinary lower-triangular partial-permutation form.
Choose an odd prime q avoiding all prepared denominators and
    all diagonal numerators needed to keep the factors invertible.
Form the fixed list of child multipliers t_i in [1, m-1].
Set tau = 1 - 246/10^9, using the certified strict moment bound.
Return the finite tables, q, tau, and t_star = 21840.

```

The corner tests are applied to the conjugated projectors in the three selected classes. The controlled-basis lemma proves that the required rational parameter choice exists, hence that this finite search terminates. Every test can be performed by rational elimination. The lower-triangular factorization normalizes every stored pivot to one. For the second class,

$$A_2 = (I_F - \Pi_t) \otimes I_V,$$

the factor $I_F \otimes K$ commutes with A_2 . Its first-to-last block after conjugation by T is diagonal in i , with entries $[G_i(I_F - \Pi_t)G_i^{-1}]_{1,h}$. These nonzero entries are normalized by fixed diagonal units. They therefore give one identity block in matching physical order between the first H_0 fields of H and the last H_0 fields of D . The Schur complement supplies the second, contiguous middle identity block. It is this prescribed matching order that permits the corner to use a single block interchange.

The common conjugation preserves every edge rank and every all-role endpoint difference I_m . Ordinary rational elimination constructs all unselected factorizations. The prime can be selected by trial division after the finite forbidden factors are known. None of this preparation depends on the input width.

For $m = 21952$ and $h = 28$, write $B = v^2 R$ and $N = v^3$. The prepared calls for the three selected classes are

class	copies	rank	singletons	contiguous block sizes
A_1 , stage 1–3 join	B	21896	56	21840
A_2 , stage 2 sink	B	21168	0	784, 20384
A_3 , stage 3 data entrance	$2N$	21141	811	20330

The singleton count, including all ordinary calls on other edges, is

$$n_1 = s - B(21840 + 20384 + 784) - 2N \cdot 20330 = 105555640096680883200.$$

Thus the child multipliers are exactly

$$1, \quad 784, \quad 21840, \quad 20384, \quad 20330,$$

with multiplicities $n_1, B, B, B, 2N$, respectively. The exact moment is

$$\Psi(x) = \frac{n_1 + B(784^x + 21840^x + 20384^x) + 2N \cdot 20330^x}{Wm^x}, \quad \Psi(1) = \frac{s}{Wm}.$$

The controlled-basis certificate proves

$$\Psi\left(1 - \frac{246}{10^9}\right) < 1.$$

All other edges retain the ordinary rank-one implementation, including any full-rank edge.

Depth and row contract. Every recursive invocation below receives an explicit row field preceding both target chunks. Its row count R is divisible by $W^{D(e)}$. Spectator fields of arbitrary lengths may occur elsewhere.

DEPTH(e) :

```

depth = 0
while e >= m:
    e = t_star * floor(e/m)
    depth = depth + 1
return depth

```

This computes the function D of Section 4.1. It terminates, is nondecreasing, and bounds the number of recursive circuit invocations on every dependency path. Every child of a width- e node has depth bound at most $D(e) - 1$.

The elementary componentwise shear. Suppose two groups each contain t contiguous width- f fields, with the first group before the second. The groups can be separated by arbitrary spectators. Addition and subtraction in the following routine are componentwise modulo q^f .

BLOCK_SHEAR(H_block, D_block, t, f, R) :

```

for i = 1, ..., t:
    D_i = D_i + H_i                # ordered-affine streaming
INTERCHANGE_WITH_ROWS(H_block, D_block, t*f, R)
for i = 1, ..., t:
    D_i = H_i - D_i                # ordered-affine streaming

```

The returned pair is $(H + D, D)$. The sole recursive call exchanges the two contiguous blocks without reversing their internal field order. For $t = 1$ the same routine implements an ordinary pivot shear, including when its two fields have different coordinate indices.

```

EDGE_SHEAR(A, H, D, f, R_role):
    Read the stored factors A = E1 * Pi * E2.
    Apply D = E2*D and H = inverse(E1)*H by ordered streaming.
    For each stored singleton pivot (i,j):
        BLOCK_SHEAR(H_i, D_j, 1, f, R_role).
    For each stored identity block (row_start, column_start, t):
        BLOCK_SHEAR(the t consecutive H fields at row_start,
                    the t consecutive D fields at column_start,
                    t, f, R_role).
    Apply H = E1*H and D = inverse(E2)*D by ordered streaming.

```

This is the address shear $(H, D) \mapsto (H + AD, D)$. The finite factors are invertible over $\mathbb{Z}/q^f\mathbb{Z}$ for all positive f . Each unbatched pivot uses one width- f call. Classes A_1, A_3 use their singleton corner calls and one middle-block call. Class A_2 uses two block calls, of widths $784f$ and $20384f$, with no singleton corner calls. All these widths are strictly less than mf .

The finite circuit on complete rows. For a main width mf , let H, D denote its two m -component field vectors. All edges below act on one role stream, whose row count is R/W and logical volume is V/W .

```

CIRCUIT_SHEAR(H, D, f, R):
    Split each complete row by its index  $u = Wg + w$ .
    Process the fixed circuit in chronological order:
        On each edge  $e$ :
            EDGE_SHEAR(A_e, H, D, f, R/W) on its role stream.
        At each scalar gate:
            Apply its fixed F_3 table pointwise at matching addresses.
    Merge roles, undoing the known final role permutation.
    Return the original row order.

```

```

MAIN_INTERCHANGE(H, D, f, R):
    Apply D = D - H componentwise by ordered-affine streaming.
    CIRCUIT_SHEAR(H, D, f, R)
    Apply D = H - D componentwise by ordered-affine streaming.

```

The all-role endpoint identity makes CIRCUIT_SHEAR equal to $H \leftarrow H + D$ on every original row, including rows assigned to auxiliary roles. The three displayed operations in MAIN_INTERCHANGE therefore exchange the two main chunks. All gates allow arbitrary initial auxiliary payloads. Pointwise operations preserve the row and spectator coordinates, and the final merge returns every row to its original index.

An integer width with already available rows.

```

INTERCHANGE_WITH_ROWS(H, D, e, R):
    Require R to be divisible by  $W^{\text{DEPTH}(e)}$ .
    if  $e < m$ :
        Interchange the  $e$  corresponding digits individually; return.
     $f = \text{floor}(e/m)$ ;  $r = e - m*f$ .

```

```

Split H = (H_plus,H_minus), D = (D_plus,D_minus),
    with the two high parts of length r.
Interchange the corresponding high-part digits individually.
Gather H_plus immediately before H_minus, obtaining
    (D_plus, H_plus, H_minus, gap, D_minus).
MAIN_INTERCHANGE(H_minus, D_minus, f, R).
Move H_plus back immediately before the final H_minus.

```

The result is $(D^+, D^-, \text{gap}, H^+, H^-)$. Fewer than a fixed multiple of m single-digit operations handle the remainder. No chunk range is padded inside this routine. Every recursive child width is at most $t_* \lfloor e/m \rfloor$, so splitting the rows once supplies its full row-divisibility requirement.

An arbitrary radix- q call without a row promise.

```

INTERCHANGE_RADIX_Q(H, D, e):
    depth = DEPTH(e)
    rho = least nonnegative integer with  $q^{(2\rho)} \geq W^{\text{depth}}$ 
    if  $e < m$  or  $2\rho \geq e$ :
        Interchange corresponding digits individually; return.
    Split the high rho digits from each chunk.
    Interchange those high parts digit by digit.
    Gather the two high parts before both remaining chunks.
    Regard their joint range  $q^{(2\rho)}$  as an opaque row field.
    Pad complete rows to  $R = W^{\text{depth}} * \text{ceil}(q^{(2\rho)}/W^{\text{depth}})$ .
    INTERCHANGE_WITH_ROWS(the remaining chunks,  $e - \rho$ , R).
    Remove the padded rows.
    Undo the gathering move, preserving the completed high-part swap.

```

The row range is enlarged by less than two, once. The existing high digits supply all its other capacity. Since $\rho = O(\log e)$, the fallback condition $2\rho \geq e$ holds for only a fixed finite set of widths. Thus its individual-digit cost changes only the constant in the asymptotic bound. Computing the displayed powers and comparisons uses ordinary integer arithmetic; no real logarithm is required at runtime. The row count is divisible by $W^{D(e)}$, hence also by $W^{D(e-\rho)}$.

Binary input chunks. For equal binary chunk widths u , choose the least integer e with $q^e \geq 2^u$. Pad each target range numerically from $[2^u]$ to $[q^e]$, putting zeros on invalid addresses. Call `INTERCHANGE_RADIX_Q`, then delete the invalid addresses. This is the original streaming radix conversion: numerical order is preserved, each target range expands by less than q , and total volume expands by less than q^2 . The completed exchange preserves the set of valid address pairs. Since q is fixed and $e = O(u)$, this gives $O(Vu^\tau)$ for arbitrary binary chunk widths and arbitrary prefix, gap, and suffix ranges.

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The bulk-recursion contribution is by icekylinx, with substantial OpenAI GPT-6 Astra assistance. Review corrections and integration used OpenAI Codex assistance. Prior constructions retain the attribution listed above and in the bibliography.

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